

BPM 1
CPR module
Lecture 30

**Respiratory Muscles and
Breathing Cycle**

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Learning Objectives

- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0199** Describe how pleural pressure, alveolar pressure, airflow, lung recoil and lung volume change during a normal quiet breathing cycle or a deep breathing cycle with forced expiration.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0200** Identify the points of onset of inspiration, cessation of inspiration, and cessation of expiration in a figure representing Intrapleural pressure, airflow and alveolar pressure changes.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0201** Describe how differences in pressure between the atmosphere and alveoli cause air to move in and out of the lungs.

Learning Objectives

- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0202** Describe pulmonary pressure-volume (compliance) curve, inflation and deflation limbs. Explain the cause and significance of the hysteresis in the curves.
- **SOM.MK.II.BPM1.3.CPR.2.PHYS.0203** Describe the factors that alter lung compliance in most common lung diseases.

Recommended Reading

Rhoades and Bell, Medical Physiology Principles for clinical medicine, 6e, Chapter 18, Ventilation and the Mechanics of Breathing.

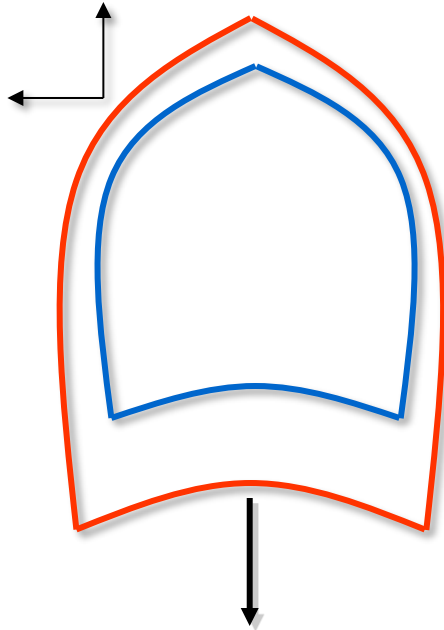
Online Link: : <https://meded-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253088352&bookid=3211>

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Muscles of Respiration

Inspiration

External Intercostals
Scalene
Sternomastoid

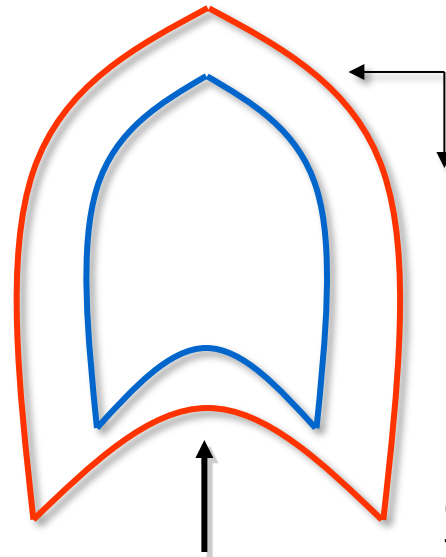


Diaphragm Contracts
shortens and pulls flat

Tidal 1 cm movement
10 cm during workload

Expiration

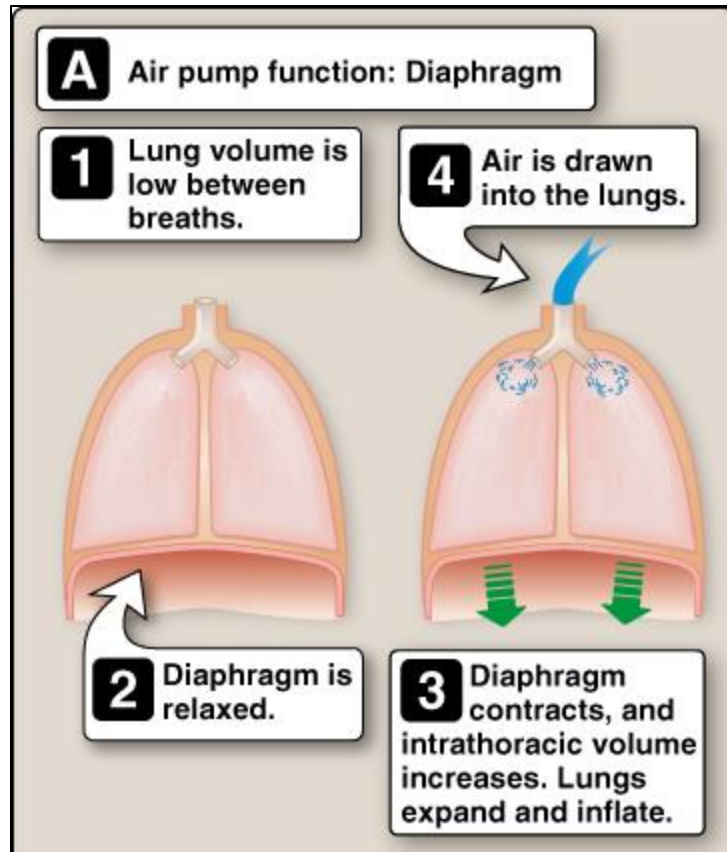
Internal Intercostals
Rectus Abdominis
Obliques
Transversus Abdominis



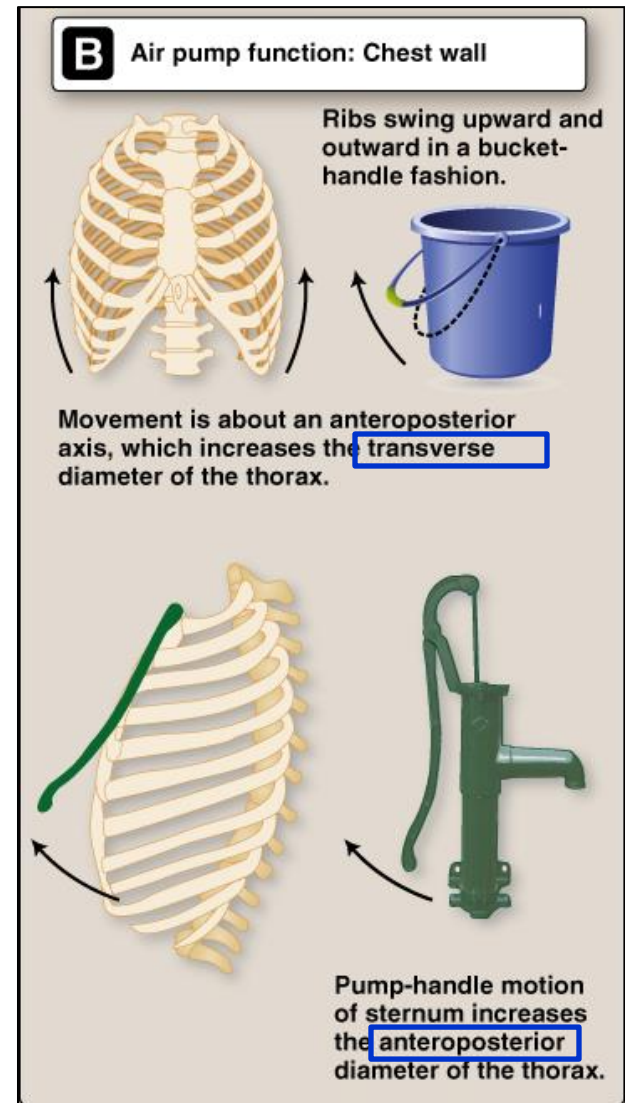
Diaphragm Relaxes

Bucket Handle Analogy:
As ribs lift upward – they swing out
Increasing volume

Thoracic volume changes during inspiration



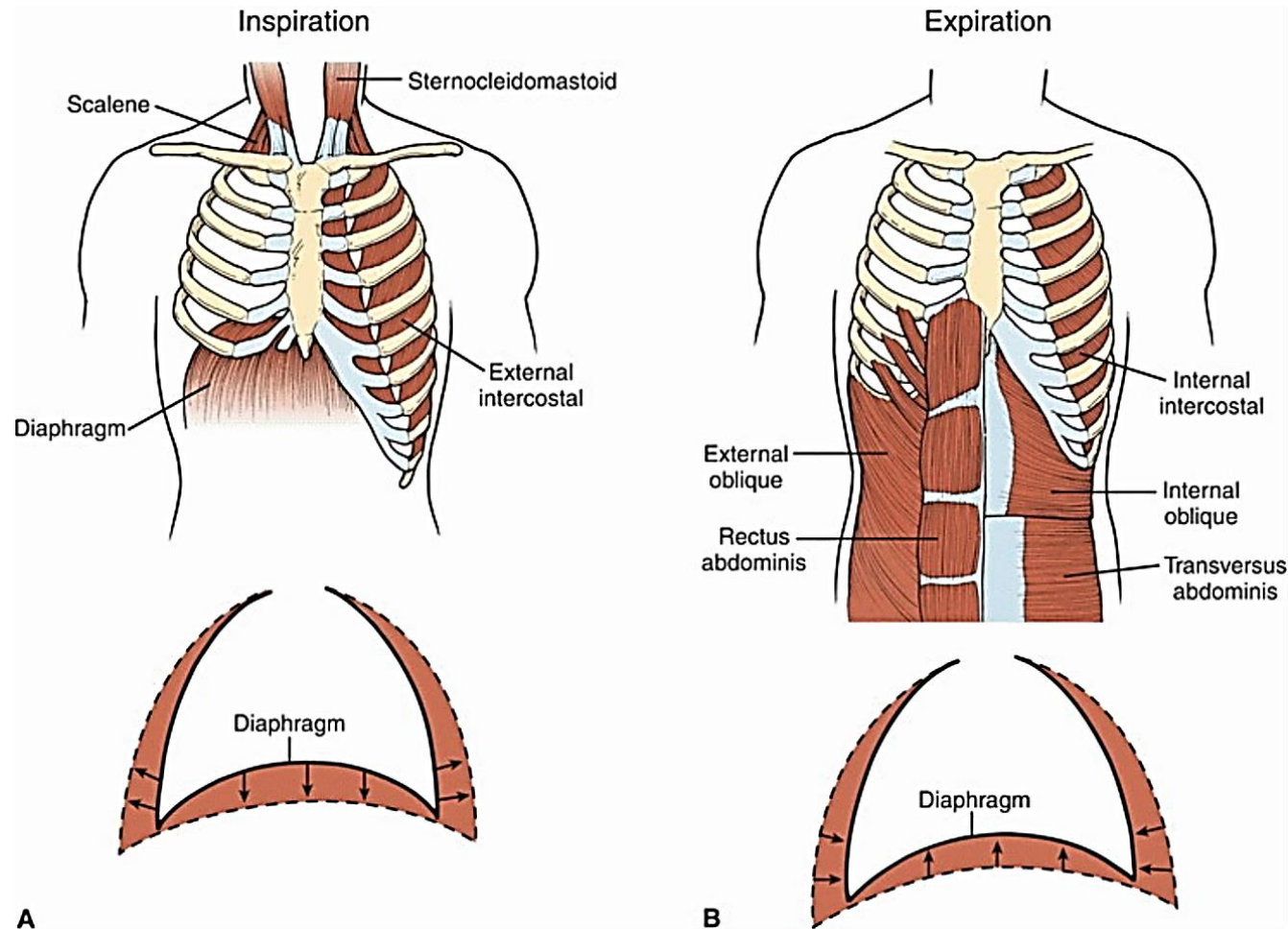
Note: Expiration is a passive process due to the relaxation of diaphragm



Explanation for the previous slide

- During normal tidal breathing, the **diaphragm** is the main muscle responsible for driving ventilation. In tidal breathing, its contraction and shortening pull it flatter and thus increase the volume inside the thoracic cavity; it enlarges the **vertical dimensions** of the chest. Relaxation and lengthening of the muscle causes it to pull back upward due to elastic recoil of the lungs.
- Contraction of the external intercostals between the ribs pull the ribs upwards like a bucket handle, and sternum upwards like a pump-handle increase the transverse and anterior-posterior dimensions of the chest, respectively. These movements also increase thoracic volume.
- During exercise, the ventilatory demands are greater and accessory muscles are recruited into the respiratory process. Stronger contraction of the external intercostals between the ribs pull the ribs upwards and outwards, thereby increase thoracic volume further.
- The scalene and sternomastoid muscle are accessory muscles that help to stabilize the first rib during forced inspiration.
- Expiration is not normally an active process, but a relaxation of the muscles involved in inspiration. However, with great ventilatory demand, expiration can become active. In this instance, the internal intercostals and the **abdominal muscles** pull the ribcage downwards and inwards, thus increasing intrapulmonary pressure and squeezing air out of the lungs.

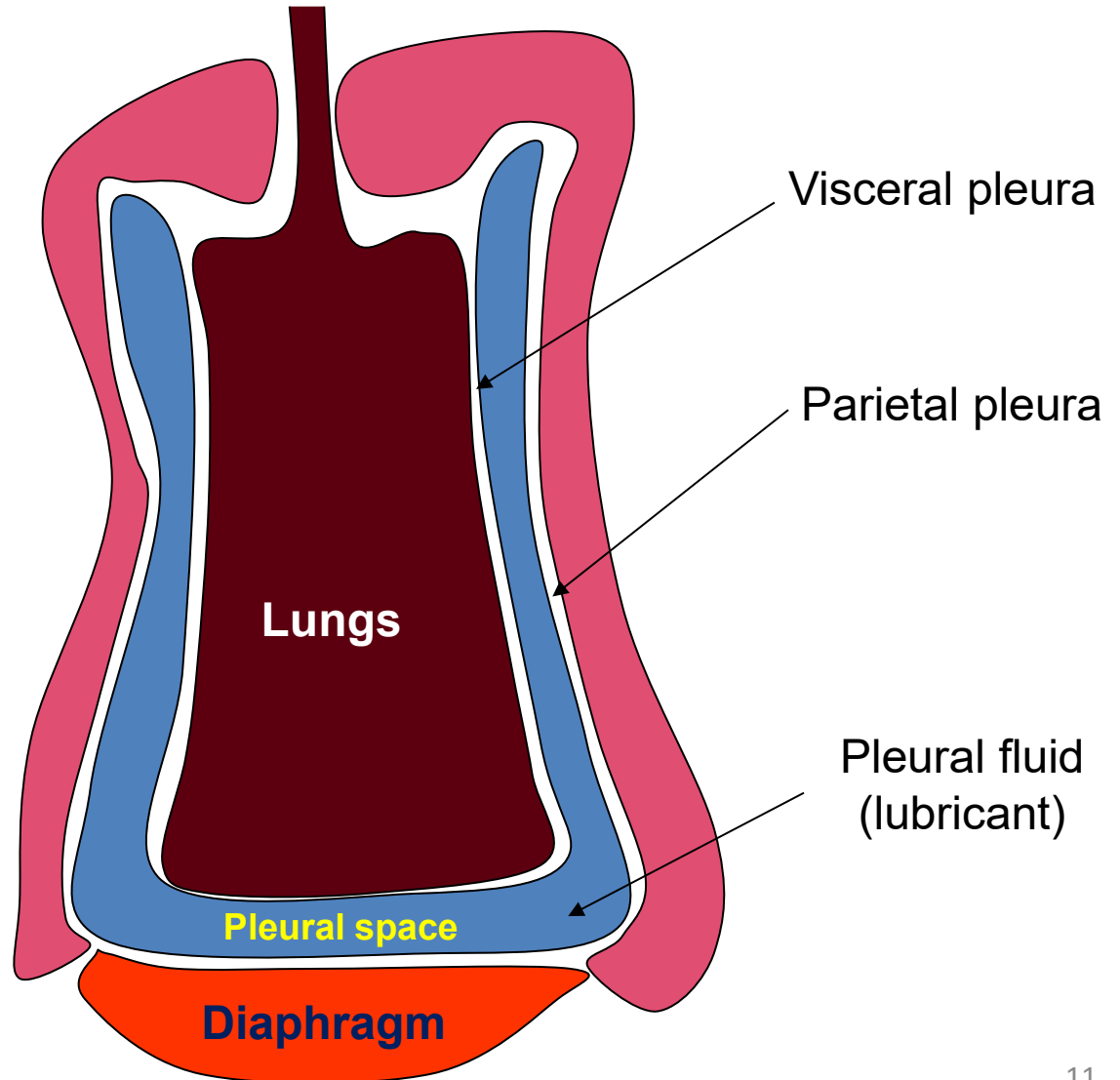
Muscles of Respiration



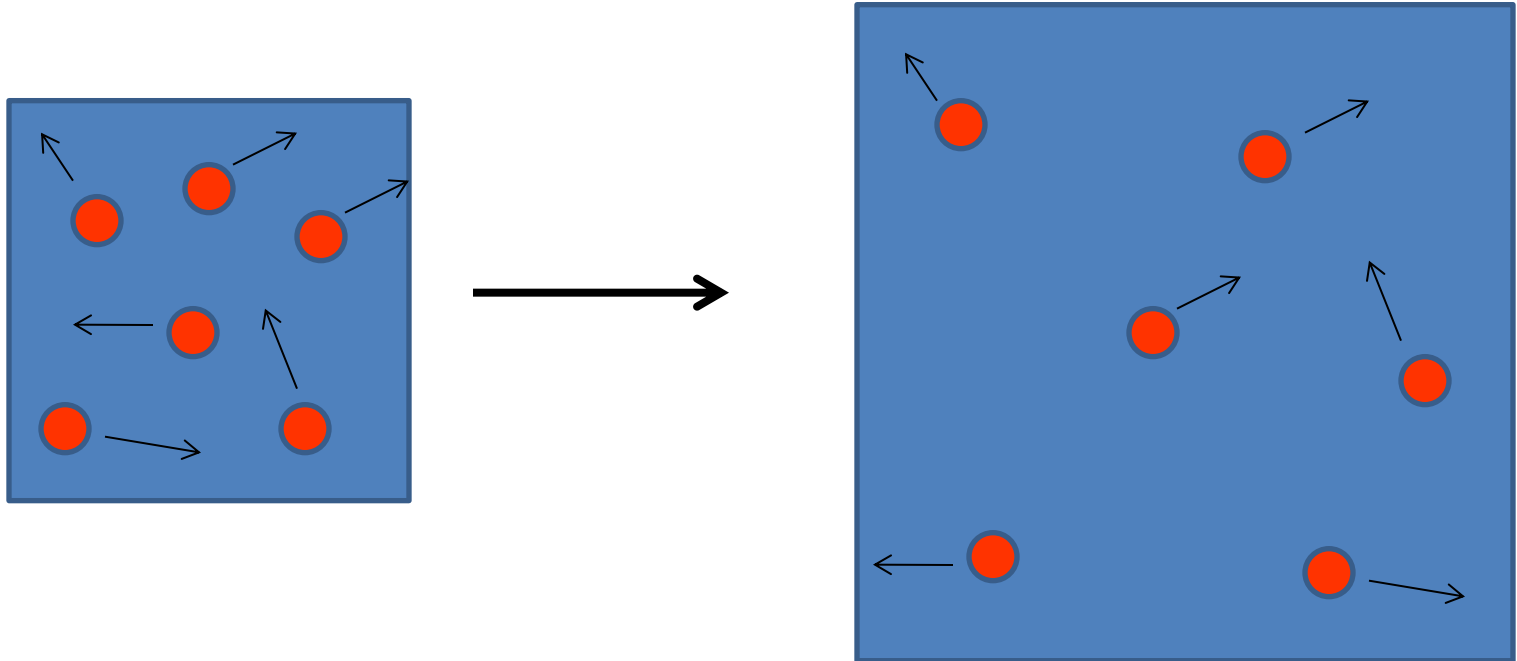
- A. During inspiration, the **diaphragm** contracts and pushes the abdominal contents downward. The downward movement also pushes the rib cage outward. With deep and heavy breathing, the accessory muscles (the **external intercostal muscles**, **sternocleidomastoids**) also contract and pull the rib cage upward and outward.
- B. Expiration is passive during resting conditions. The diaphragm relaxes and returns to its dome shape, and the rib cage is lowered. During forced expiration, the **internal intercostal muscles** contract and pull the rib cage downward and inward. The **abdominal muscles** also contract and help pull the rib cage downward, compressing thoracic volume.

Relation between Pleura, Lungs and Chest wall

- Visceral pleura is physically attached to the lung surface.
- Parietal pleura is attached to the chest wall and the diaphragm.
- The chest wall pulling out and the lungs pulling inwards create a **negative pressure in the pleural space**.



Concept of Pressure and Volume



In a closed system,
Increasing Volume of a given mass of gas **drops Pressure**

Combined Gas Law

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

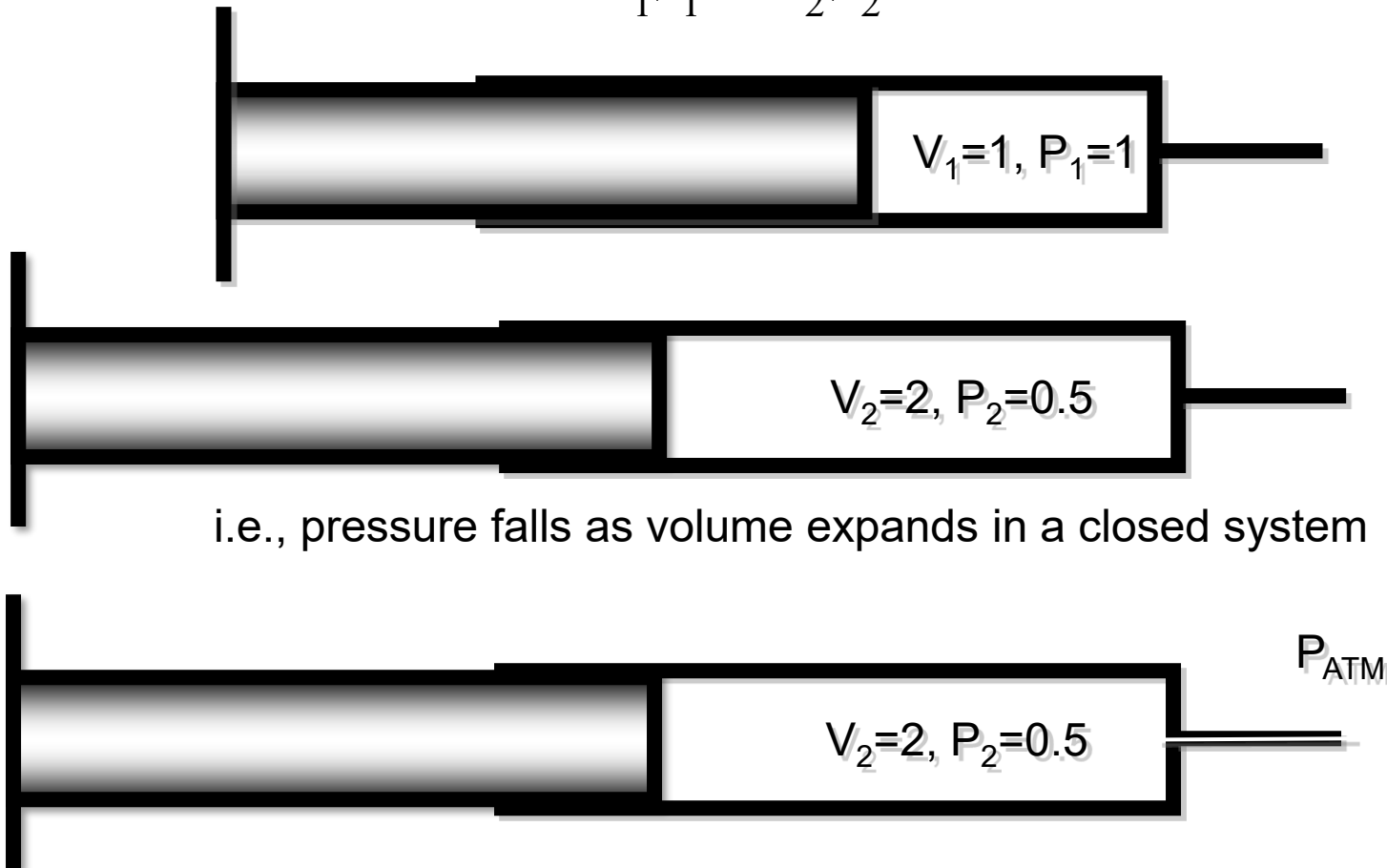
In a closed system (when T remains constant),

**Increasing Volume will cause a Pressure drop;
Decreasing Volume will cause a Pressure rise.**

P = pressure; V = volume; T = temperature

Robert Boyle's Law

$$P_1 V_1 = P_2 V_2$$

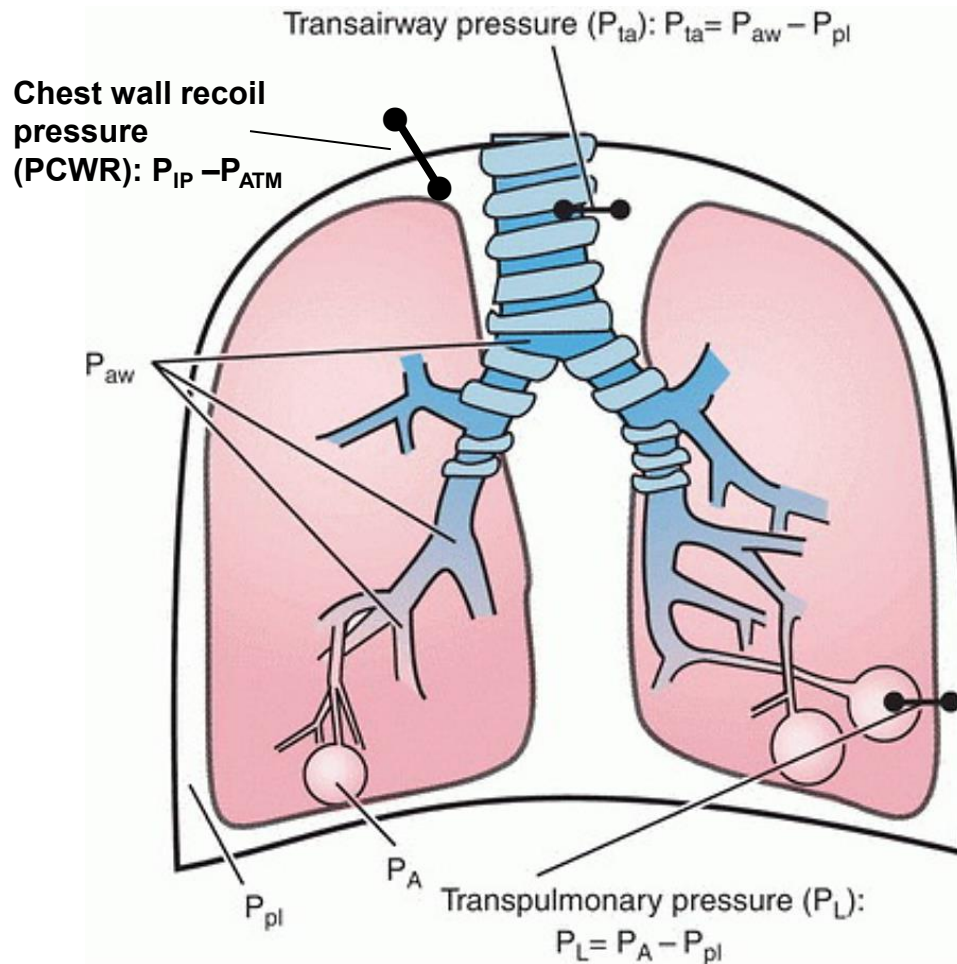


i.e., pressure falls as volume expands in a closed system

Open the end of the syringe now – what happens?

$P_2 < P_{ATM}$ = airflow into syringe until $P_2 = P_{ATM}$

Pressures and Gradients in the thorax



Important pressures involved in breathing:

Airway pressure (P_{aw}),
Alveolar pressure (P_A), and
Pleural pressure (P_{pl}).

Pressure Gradients:

Defined as the **pressure inside minus the pressure outside**.

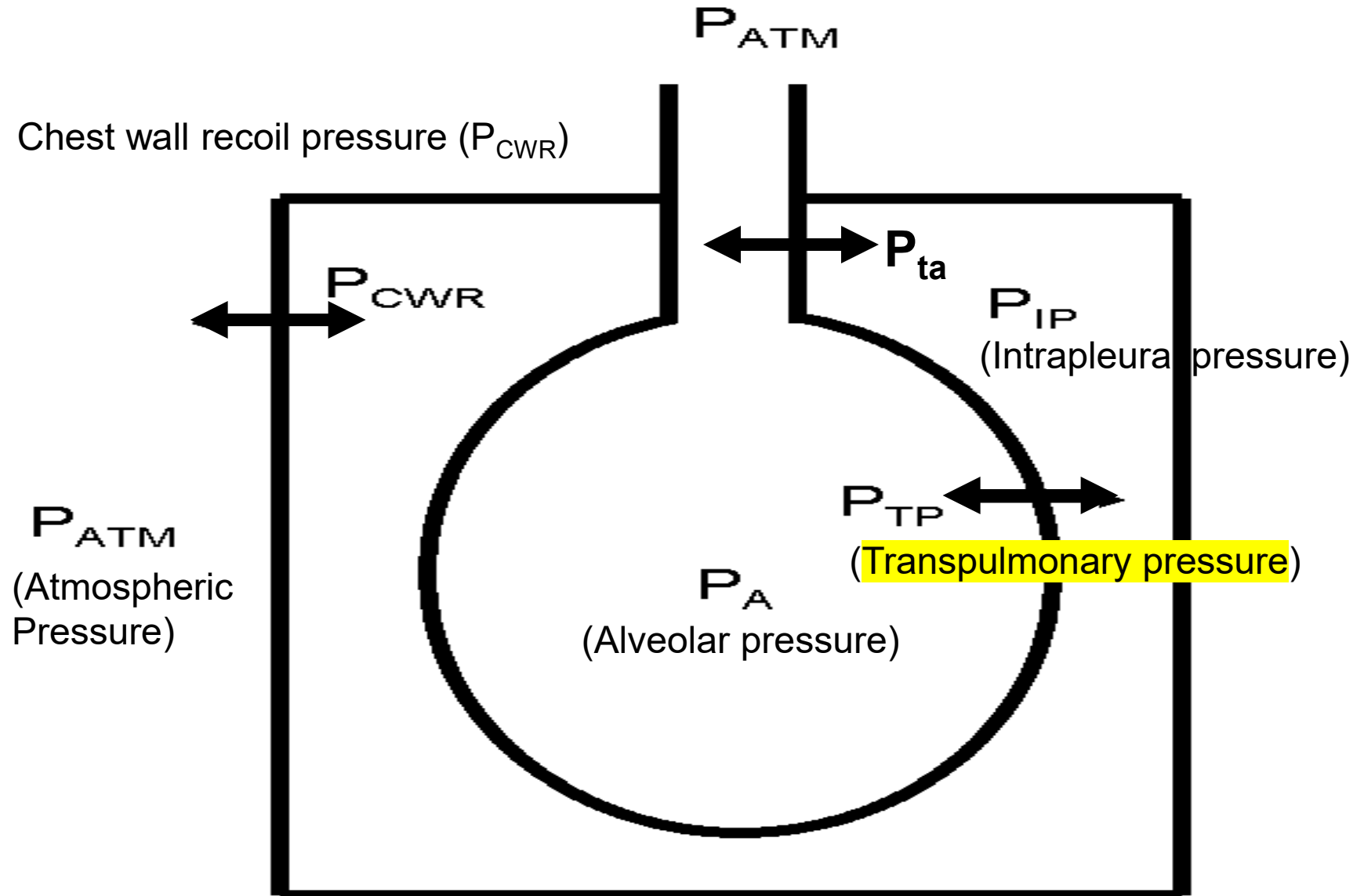
3 different pressure gradients are:

Transpulmonary pressure (P_L),

Chest wall recoil pressure (PCWR), and

Transairway pressure (P_{ta}).

Pressures and Gradients



Respiratory Pressures and Pressure gradients

Important Respiratory Pressures and Pressure gradients:

1. Intrapleural pressure (P_{IP}) = pressure inside pleural space.
 2. Alveolar pressure (P_A) = pressure inside alveolus.
 3. Transpulmonary pressure (P_{TP}) = the transmural pressure across the alveolar wall ($P_A - P_{IP}$).
- (Transmural pressure means any pressure gradient across a wall – e.g., airway wall or alveolar wall)

The box in the previous slide represents the chest wall, and the big oval shape inside the box represents the alveolar space of the lung.

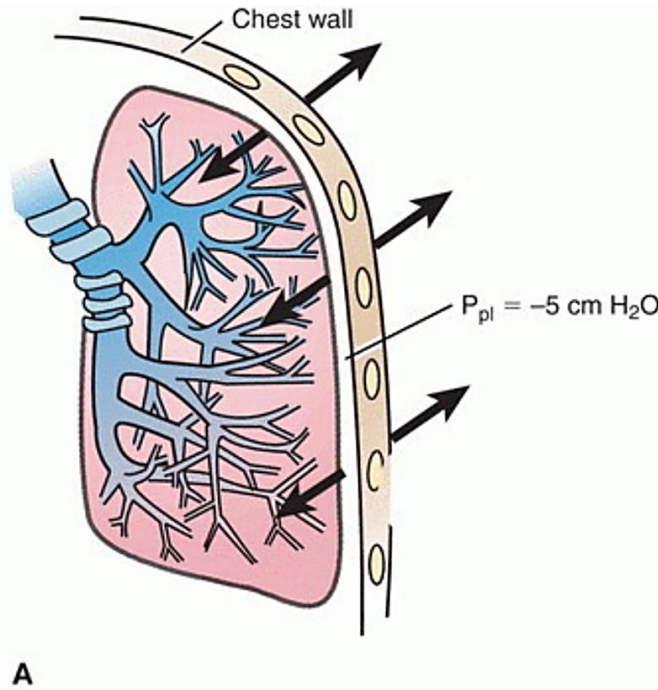
Importance of Pressure Gradients:

Airflow into the lungs is generated by $P_A - P_{ATM}$ gradient.

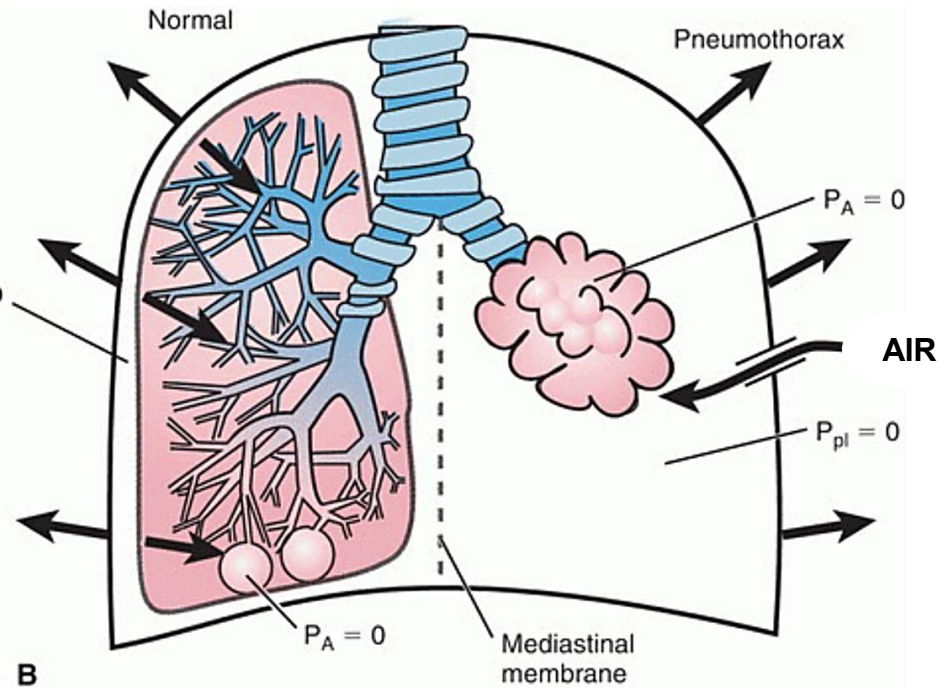
Transpulmonary Pressure $P_{TP} = P_A - P_{IP}$ (also called lung elastic recoil pressure) helps to keep alveoli (which tries to collapse) open.

Chest wall recoil pressure PCWR = $P_{IP} - P_{ATM}$ keeps the chest wall as a tight spring which tries to expand outwards.

Intrapleural pressure: Features



(A) The stretched lung (at the end of a quiet expiration)

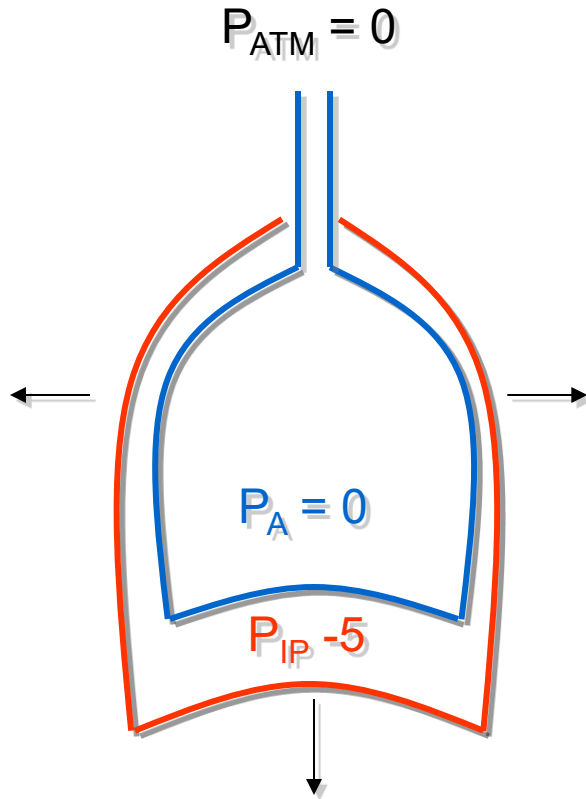


(B) Pneumothorax in the left side chest

Intrapleural pressure: Features

- Intrapleural pressure is **always negative** (i.e., less than atmospheric pressure) during quiet breathing.
- This is because of 2 opposite forces pulling the lung and chest wall in opposite directions.
- The stretched lung (at the end of a normal inspiration or expiration) tends to recoil inwards, and the chest wall tends to recoil outwards, but in equal and opposite directions.
- A puncture of the lung or chest wall results in suction of air into pleural space and it is called **pneumothorax**.
- In pneumothorax, the transpulmonary pressure becomes zero and elastic recoil causes the lung to collapse. The mediastinal membrane prevents the other lung from collapsing.
- In pneumothorax, intrapleural pressure becomes positive, either equal to atmospheric pressure or greater than it.

Pressure Changes during Breathing



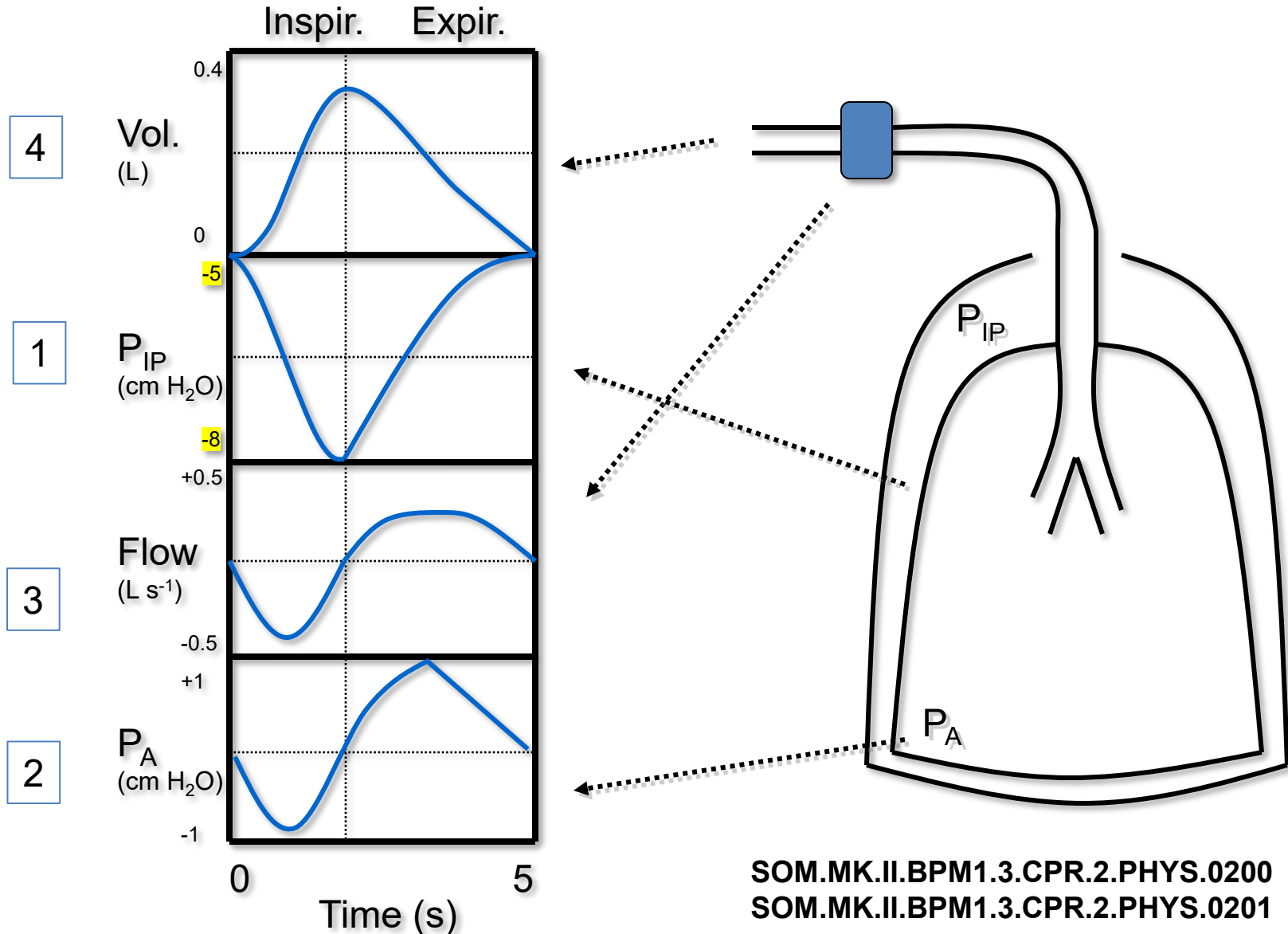
1. Inspiratory muscles contract
2. Thoracic cavity expands in volume
3. Pleural pressure falls (from -5 to -8)
4. Transpulmonary pressure gradient increases $P_{TP} = P_A - P_{IP}$
5. $\uparrow P_{TP}$ causes alveolar expansion whilst $\downarrow P_{TP}$ causes alveolar recoil
6. Alveoli expand in volume
7. Alveolar pressure falls (from 0 to -1)
8. Air flows down the gradient from P_{ATM} to P_A

Explanation for the previous slide

The sequence of events that lead to inspiration are:

1. The diaphragm contracts and increases the volume of the thoracic cavity.
2. The pleural pressure (which is –ve) becomes even more negative due to the downward displacement of the diaphragm (like going from a weak vacuum to a strong vacuum).
3. This creates a pressure gradient between the pleural space and the alveolar space and a greater transpulmonary pressure.
4. The alveoli expand due to the increase in transpulmonary pressure.
5. The increase in volume of the alveolar space leads to a drop in the alveolar pressure.
6. A pressure gradient now exists between the alveolar pressure and the atmospheric pressure in the mouth.
7. Air flows down this pressure gradient. As air flows, the pressure gradient diminishes. Air flow continues until there is no pressure gradient between the atmosphere and the alveoli. At the end of inspiration, alveolar pressure becomes equal to the atmospheric pressure.

Breathing Cycle: Changes in pressure, Volume & Flow



Explanation for the previous slide

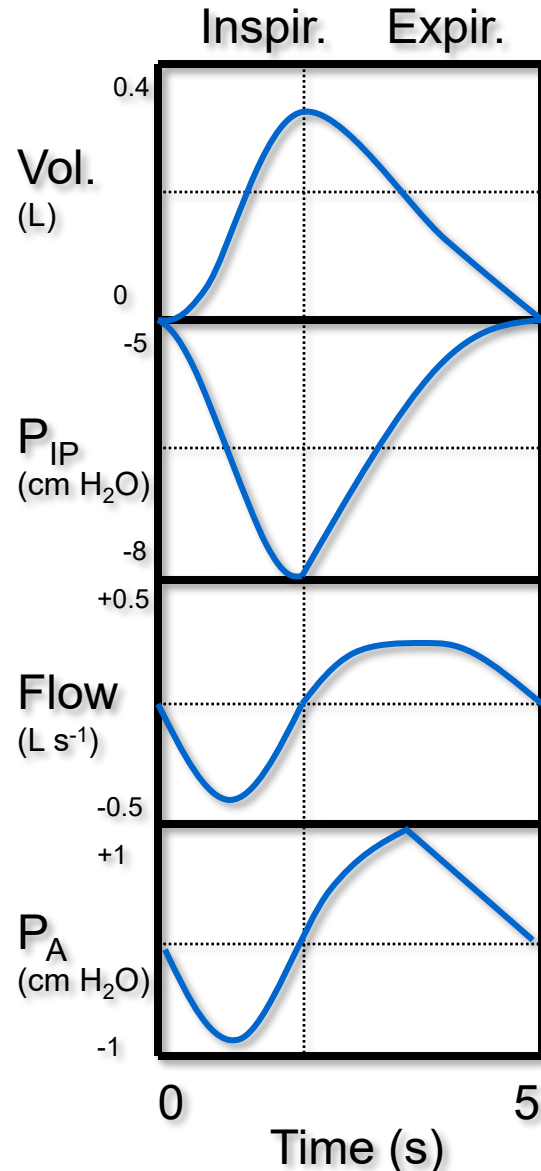
This slide demonstrates the changes in pleural pressure, alveolar pressure, lung volume and air flow over normal tidal breathing cycle.

The first thing to notice is that inspiration is slightly shorter than expiration. Reasons for this are: Inspiration is an active process and **airway resistance** is higher during expiration, therefore airflow is slower.

During inspiration: At time zero in fig, the pleural pressure falls from **-5 to -8 cm H₂O**, due to expansion of the thoracic volume when the diaphragm contracts. Alveoli expand at the same time, therefore, alveolar pressure falls. As soon as alveolar pressure starts to fall, airflows in. The initial air flow rate is high, but as the air flow into the lungs reduces the pressure gradient between the atmosphere and the alveolar space, the airflow rate reduces. Air flow continues until the pressure gradient becomes zero towards the end of the inspiration.

During expiration: The diaphragm relaxes and the thoracic volume decreases. Elastic recoil of the lung causes the alveoli to diminish in size and the pleural pressure to become less negative. This squeezes the air in the lungs, increasing alveolar pressure and creating a pressure gradient, favoring airflow out of the lungs.

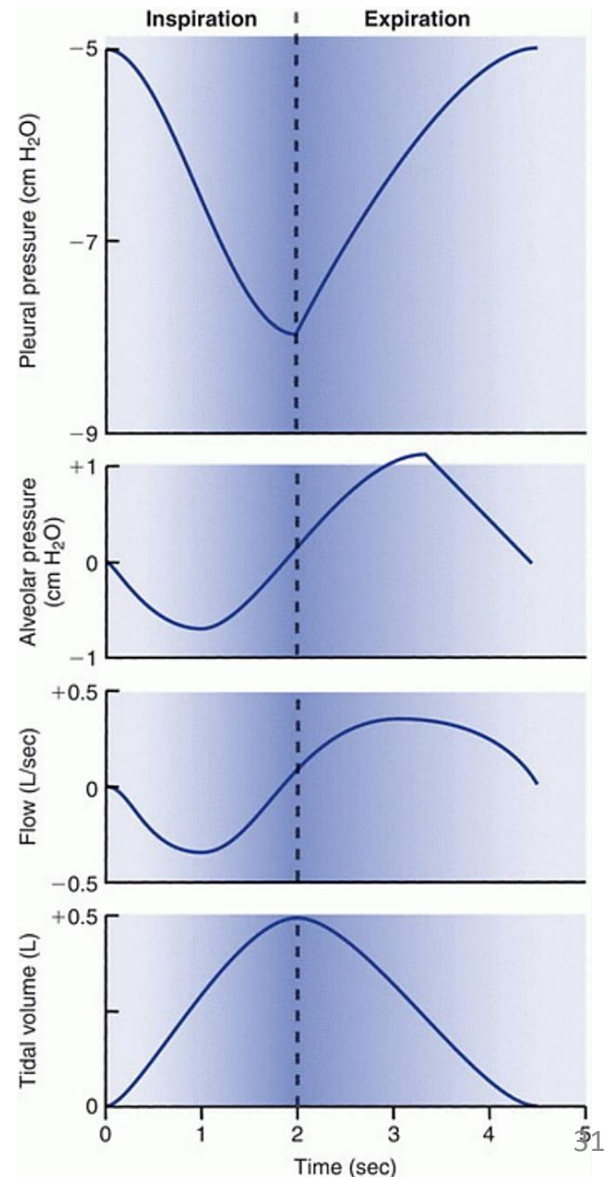
Breathing Cycle: Changes in pressure, Volume & Flow



- Inspiration at 2 sec is slightly quicker than expiration (3 sec)
- Only a small pressure change in the alveoli is enough to drive sufficient flow during normal tidal breathing
- Why is expiration slower than inspiration and why is a greater dynamic P_A required to drive out air during expiration than inspiration?
- Increased airway resistance during expiration (due to less radial traction and dynamic airway compression) creates larger pressure gradient but slower dissipation and flow.

Why is expiration slower than inspiration

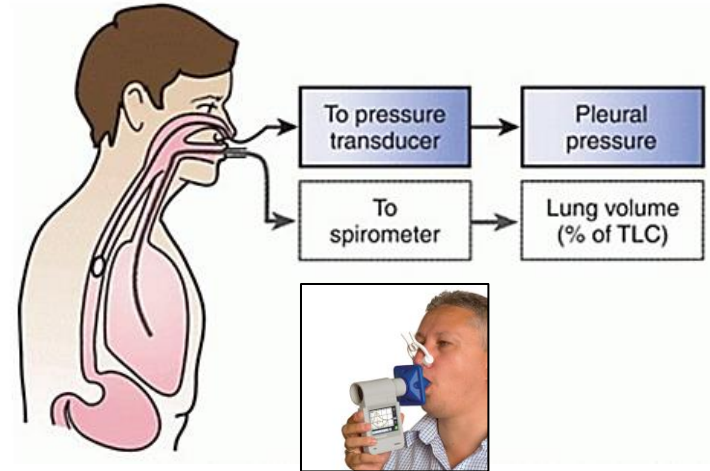
- Airflow resistance affects expiratory time. At rest, the inspiratory time (T_i) is 2 seconds and the expiratory time (T_e) is 3 seconds.
- This difference is a result of, in part, a **higher airflow resistance during expiration**, as is reflected by a higher alveolar pressure (P_A) change during expiration (1.2 cm H₂O) than during inspiration (0.8 cm H₂O).
- An increase in airway resistance decreases the T_i/T_e ratio and results in prolongation of expiration in patients.
- Only a small pressure difference between the mouth and alveoli is enough for a normal tidal volume to exhale.



Lung Compliance: Definition and Measurement

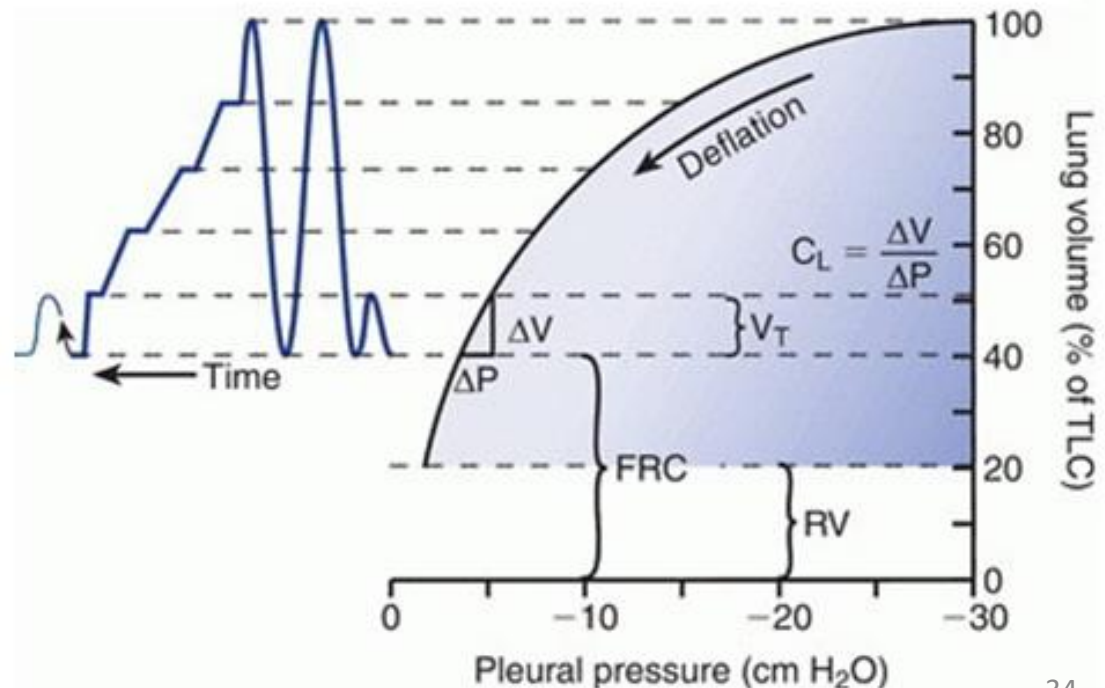
The subject first inspires maximally to total lung capacity (TLC) and then expires slowly, while airflow is periodically stopped to simultaneously measure pleural pressure and lung volume.

Lung compliance (C_L) is measured in L/cm H₂O.



$$C_L = \frac{\text{Change in lung vol}}{\text{Change in pressure}}$$

$$C_L = \frac{\Delta V}{\Delta P}$$



Significance of a low or high lung compliance

Low lung compliance indicates a stiff lung.

More work is required to inflate this lung.

Examples:

Restrictive lung diseases like pulmonary fibrosis.

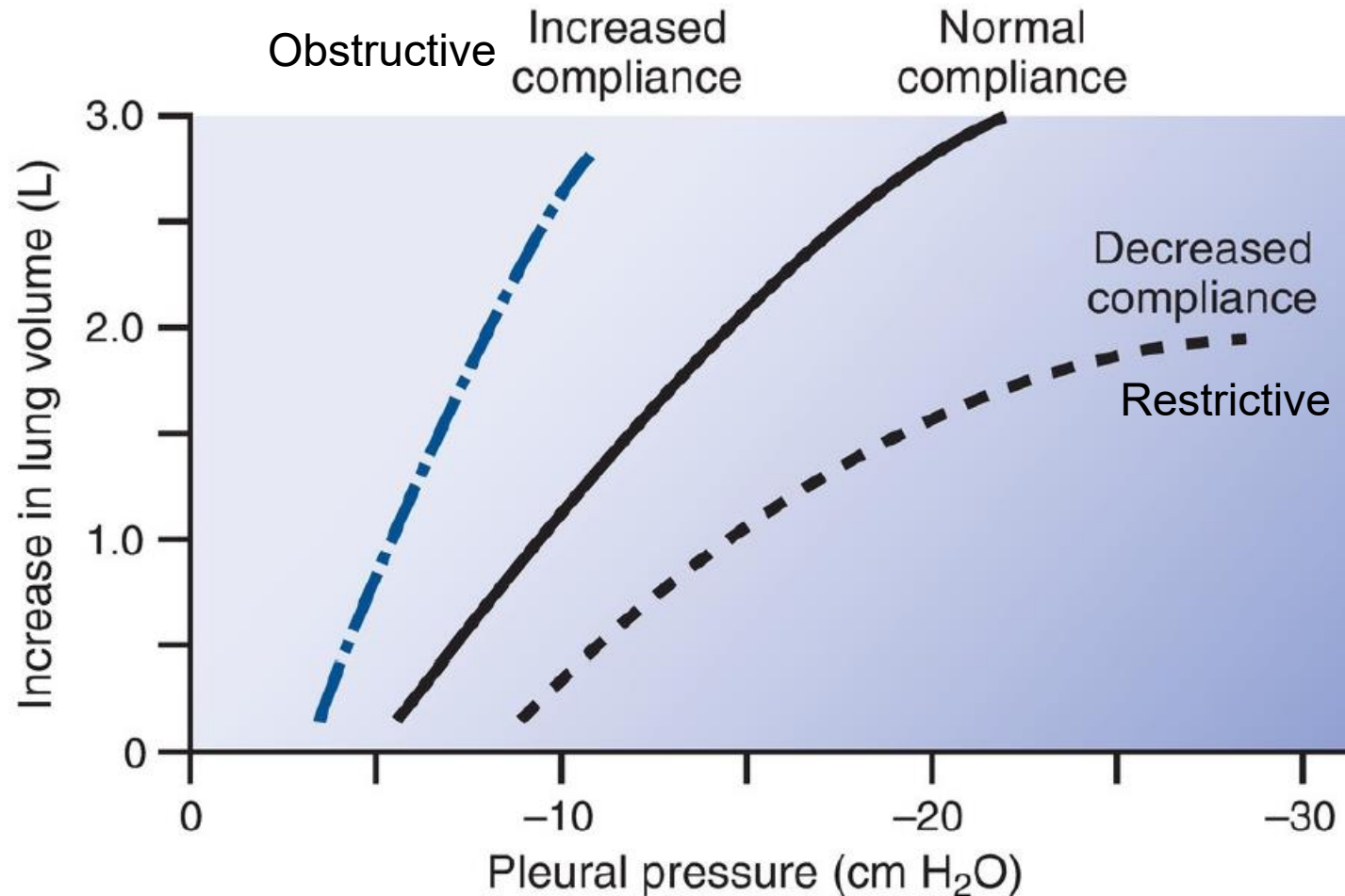
High lung compliance indicates an easily distensible lung.

These lungs have lost elastic recoil, and therefore, additional effort is required to force air out of the lungs.

Examples:

Obstructive lung diseases like Emphysema.

Significance of a low or high lung compliance



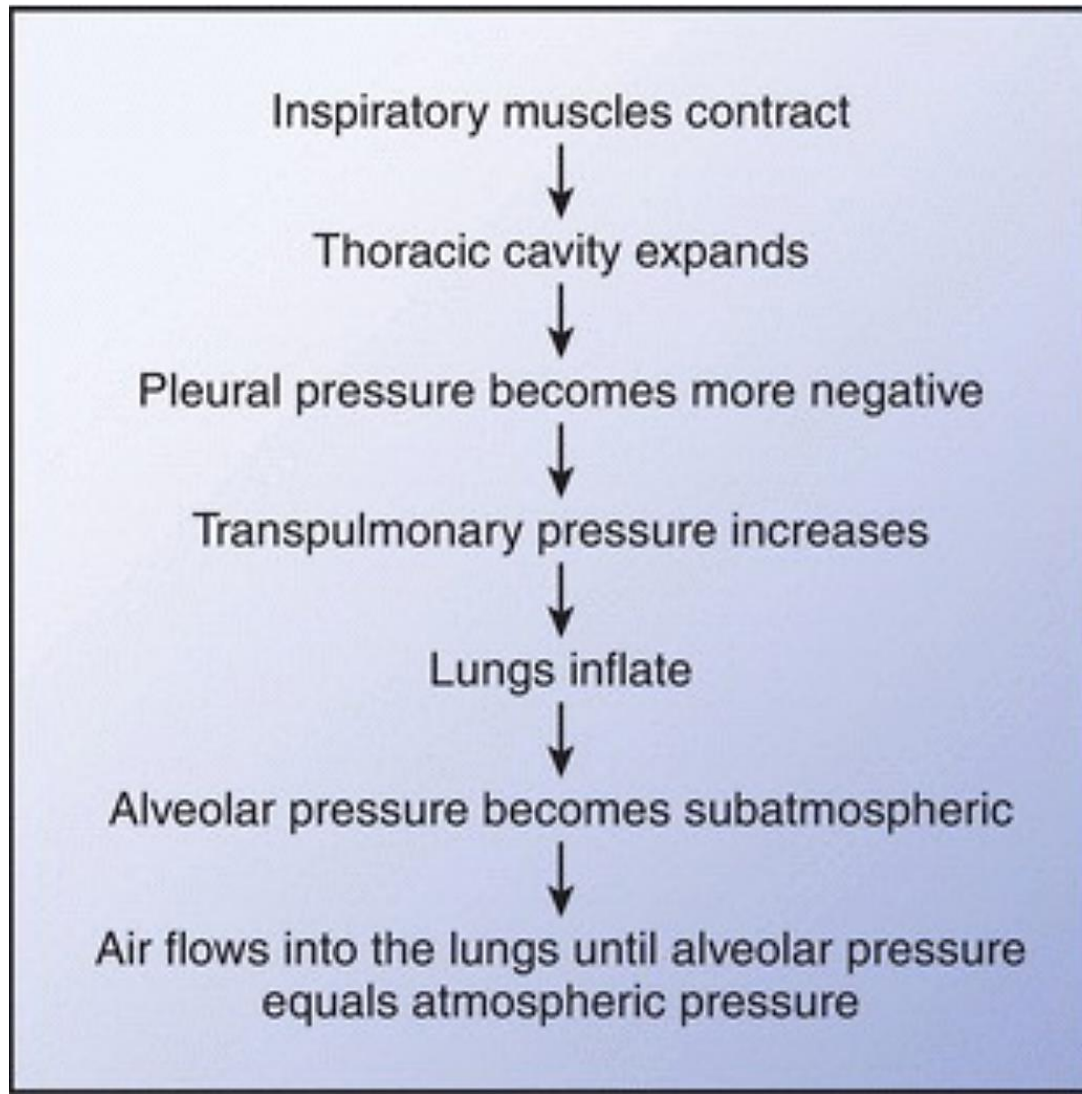
What is the significance of an abnormally low or abnormally high compliance?

Lung compliance is inversely related to lung stiffness ($C_L = 1/\text{stiffness}$).

Abnormally low lung compliance indicates a stiff lung, which means more work is required to inflate the lungs to bring in a normal tidal volume. An example of abnormally low compliance is seen in a restrictive disorder, a disease state that is characterized by restriction of lung inflation (stiff lungs). The compliance of these lungs is decreased compared with that of a normal lung. Restrictive disorders can be caused by infection or by toxic environmental insults in which the elastic properties are altered.

An abnormally high compliance is also detrimental. Lungs with abnormally high compliance have low elastic recoil, and therefore extremely easy to inflate. However, additional effort is required to force air out of these lungs. An example is emphysema, a disease in which lungs have been overstretched due to loss of elasticity. Emphysema belongs to a broader category termed chronic obstructive pulmonary disease (COPD), which is a diseased state defined as an obstruction of airflow out of the lungs. Lungs with increased compliance have high static lung volumes (increased TLC, FRC, and RV) and low airflow. Therefore, patients with COPD have high lung volumes and retain an abnormally high RV of air.

SUMMARY



Airflow parallels alveolar pressure during breathing. The sequence during inspiration results in a fall in alveolar pressure causing air to flow into the lungs.

Any Questions ?

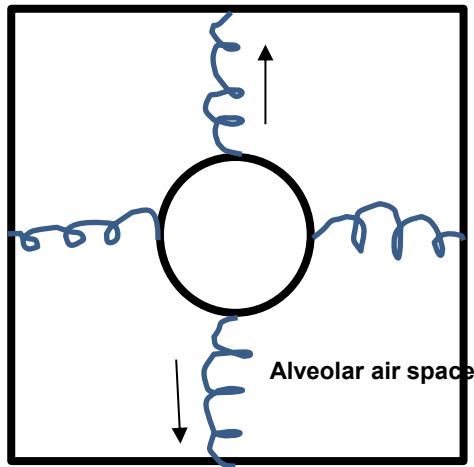
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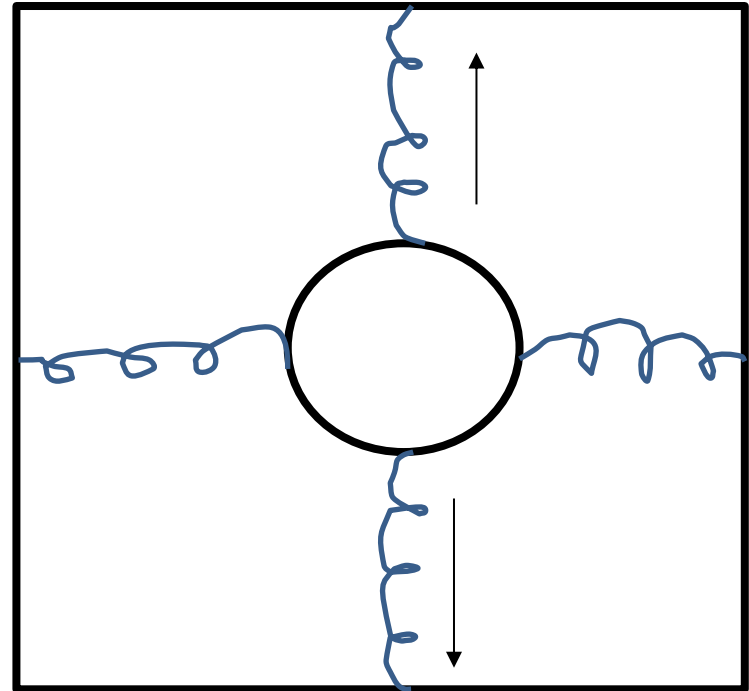


Radial Traction/Tension

Expiration



Inspiration

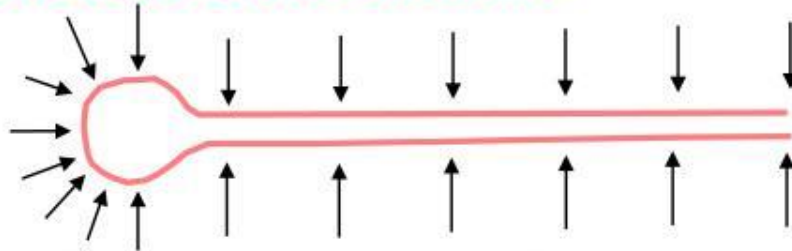


Frame with springs attached to elastic circular structure in the center. The area bounded by the springs is equivalent to the alveolar air space and the springs are equivalent to the alveolar walls.

If you expand frame (increase thorax volume) – stretches springs, they experience more tension (radial traction) and pull on the circular structure at the center. Opening it up to a larger radius

Dynamic Airway Compression

During inspiration the airways are pulled open by the expanding thorax. Therefore in cases of **increased airway resistance** expiration tends to be more difficult.



Transairway Pressure = Airway Pressure – Pleural pressure

The transairway pressure tends to compress airways during active expiration - **pleural pressure rises during expiration** (increases airway resistance)

If **no obstruction**: the increased airway resistance causes an **increase in airway pressure** upstream. This helps open the airways (i.e. **reduce the compressive transairway pressure**)

If **there is an obstruction** (e.g. COPD), the driving pressure between the alveolus and airway is lost over the obstructed segment. This causes a **fall in airway pressure** along the airways resulting in **airway compression by the transairway pressure** during active expiration.

